EXPERIMENT 1: TITANIC DATASET PREPROCESSING AND VISUALIZATION

AIM:

To perform data preprocessing and exploratory data analysis on the Titanic dataset by inspecting data, handling missing values and duplicates, encoding categorical data, scaling numerical data, and visualizing feature relationships.

ALGORITHM:

1. Import the required Python libraries: Pandas, Seaborn, Matplotlib, and scikit-learn preprocessing tools.
2. Load the Titanic CSV dataset into a Pandas DataFrame and inspect the first rows, data types, and missing values.
3. Fill missing Cabin values with “Unknown” and remove duplicate records.
4. Encode the Sex column using LabelEncoder.
5. Standardize the Fare column using StandardScaler.
6. Fill missing Age values using forward fill and backward fill.
7. Plot selected features (Pclass, Sex, Age, SibSp) using a pair plot.
8. Calculate and display the correlation heatmap for Pclass, Age, SibSp, Parch, and Fare.

1. Load the Titanic dataset and convert it into a DataFrame.

```
#EXP_NO_1
#MUTHURAJ_241801176

import numpy as np
import pandas as pd

df = pd.read_csv('titanic.csv')
data = pd.DataFrame(df)
df.head()
```

	Survived	Pclass	Name	Sex	Age	Siblings/Spouses Aboard	Parents/Children Aboard	Fare
0	0	3	Mr. Owen Harris Braund	male	22.0	1	0	7.2500
1	1	1	Mrs. John Bradley (Florence Briggs Thayer) Cum...	female	38.0	1	0	71.2833
2	1	3	Miss. Laina Heikkinen	female	26.0	0	0	7.9250
3	1	1	Mrs. Jacques Heath (Lily May Peel) Futrelle	female	35.0	1	0	53.1000
4	0	3	Mr. William Henry Allen	male	35.0	0	0	8.0500

Next steps: Generate code with df | New interactive sheet

2. Explore and Understand the Dataset

- Display the first few rows.
- Get information about column data types and missing values.

```
#EXP_NO_1
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df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 887 entries, 0 to 886
Data columns (total 8 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Survived                 887 non-null    int64
 1   Pclass                   887 non-null    int64
 2   Name                     887 non-null    object
 3   Sex                      887 non-null    object
 4   Age                      887 non-null    float64
 5   Siblings/Spouses Aboard  887 non-null    int64
 6   Parents/Children Aboard  887 non-null    int64
 7   Fare                     887 non-null    float64
dtypes: float64(2), int64(4), object(2)
memory usage: 55.6+ KB
```

3. Fill missing values in Age using Forward Fill and Backward Fill

```
#EXP_NO_1
#MUTHURAJ_241801176

df["Age"] = df["Age"].ffill()
df["Age"] = df["Age"].bfill()
df.head(10)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	35.0	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C

4. Fill missing values in Cabin with "Unknown" (limit = 5)

```
#EXP_NO_1
#MUTHURAJ_241801176
df["Cabin"] = df["Cabin"].fillna("Unknown", limit = 5)
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Unknown	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	Unknown	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	Unknown	S

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5. Remove Duplicate Records if any

```
#EXP_NO_1
#MUTHURAJ_241801176
df.drop_duplicates(inplace=True)
df.head(10)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Unknown	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	Unknown	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	Unknown	S
5	6	0	3	Moran, Mr. James	male	35.0	0	0	330877	8.4583	Unknown	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	Unknown	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	Unknown	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	Unknown	C

Next steps: Generate code with df | New interactive sheet

6. Encode the "Sex" column using LabelEncoder

```
#EXP_NO_1
#MUTHURAJ_241801176
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df["Sex"] = le.fit_transform(df["Sex"])
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	Unknown	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	Unknown	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	Unknown	S

Next steps: Generate code with df | New interactive sheet

7. Scale the Fare column using StandardScaler

```
#EXP_NO_1
#MUTHURAJ_241801176
from sklearn.preprocessing import StandardScaler
s = StandardScaler()
df["Fare"] = s.fit_transform(df[["Fare"]])
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	-0.502445	Unknown	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	0.786845	C85	C
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	-0.488854	Unknown	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	0.420730	C123	S
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	-0.486337	Unknown	S

Next steps: Generate code with df | New interactive sheet

9. Display the Correlation Heatmap for 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare'

```
#EXP_NO_1
#MUTHURAJ_241801176

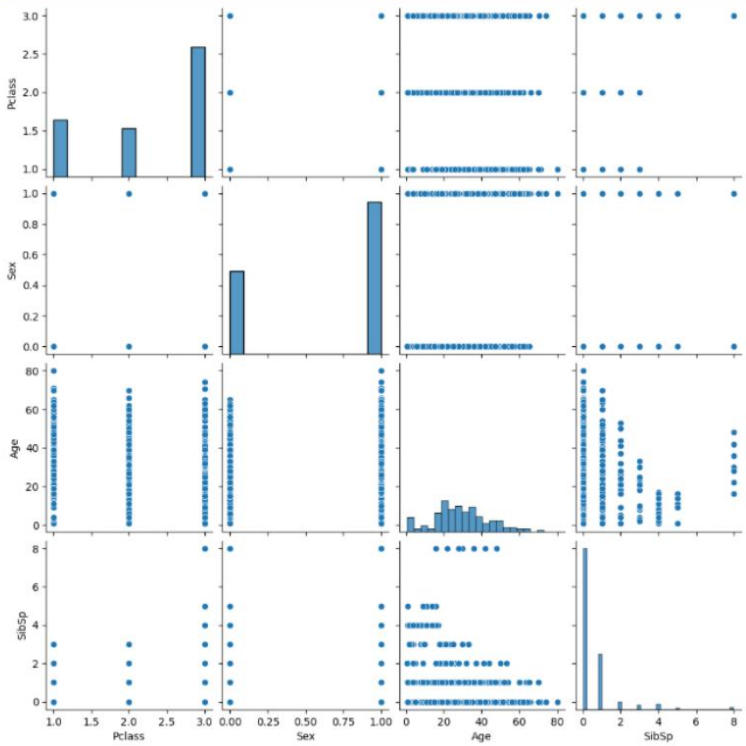
corr = df[["Pclass", "Age", "SibSp", "Parch", "Fare"]].corr()
sns.heatmap(corr, annot = True, cmap = "coolwarm")
plt.show()
```



8. Pair Plot of Selected Features such as 'Pclass', 'Sex', 'Age', 'SibSp'

```
#EXP_NO_1
#MUTHURAJ_241801176
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df[["Pclass", "Sex", "Age", "SibSp"]])
plt.show()
```



RESULT:

The Titanic dataset was loaded into a DataFrame and explored. Missing values were handled, duplicate records were removed, the Sex column was label-encoded, the Fare column was standardized, and visualizations (pair plot and correlation heatmap) were generated to examine feature distributions and relationships.